

## **EVOLOCUMAB -> cholesteryl esters to total lipids in medium VLDL percentage**

For clinical-collaborator outreach.

---

### ***Why this hypothesis is real***

Drug-target convergence on the disease scores **0.77** (noisy-OR over OT target-disease associations). This pair is fully novel relative to all known drug-indication assignments in ChEMBL. Therapeutic direction is **aligned** with OT genetic evidence (the drug pushes the target the disease-protective way). Reactome co-membership reinforces the indirect mechanism with **1** shared specific pathways.

### ***Why now (IP runway + literature footprint)***

The substance is a biologic or otherwise outside the FDA Orange Book; check the Purple Book and direct registry for IP runway.

### ***Clinical opportunity***

US prevalence figures are not in the curated map; for an orphan condition this is expected. The KOL can size the addressable population.

### ***What we propose***

**BioBix** (Department of Mathematical Modelling, Statistics and Bioinformatics, Ghent University) brings bioinformatics support to the collaboration:

- Target validation across blood / immune / cancer multi-omics datasets
- Biomarker candidate identification from public + private cohorts
- Patient stratification analytics for trial-enrichment design
- Mechanism-of-action mining via systems-biology priors
- Quick-turn evidence syntheses for IIT and partnered programs

Proposed collaboration model:

- Investigator-initiated trial framework with the substance owner
- BioBix as bioinformatics analytics partner (target/biomarker/biostatistics)
- Joint publication strategy; data-sharing agreement gated on IP terms

### ***Next step***

A 30-minute scoping call to align on feasibility. We would come prepared with a brief mechanistic dossier, an annotated literature pull, and a pilot biomarker shortlist for the disease.

**Wim Van Criekeinghe**, Professor (PhD, h-index 73)

Department of Mathematical Modelling, Statistics and Bioinformatics, Ghent University

Wim.VanCriekeinghe@UGent.be